

**AI-POWERED REAL-TIME POTHOLE DETECTION AND HAZARD
RISK ASSESSMENT SYSTEM FOR MICRO-MOBILITY (BICYCLE AND
ELECTRIC SCOOTER) USERS**

MUHAMMAD AL IEQBAL NUR HAQIEMI BIN ABDUL KHALID

UNIVERSITI TEKNIKAL MALAYSIA MELAKA

BORANG PENGESAHAN STATUS LAPORAN

JUDUL: [AI-POWERED REAL-TIME POTHOLE DETECTION AND HAZARD RISK ASSESSMENT SYSTEM FOR MICRO-MOBILITY (BICYCLE AND ELECTRIC SCOOTER) USERS]

SESI PENGAJIAN: [2026 / 2027]

Saya: Muhammad Al Ieqbal Nur Haqiemi bin Abdul Khalid

mengaku membenarkan tesis Projek Sarjana Muda ini disimpan di Perpustakaan Universiti Teknikal Malaysia Melaka dengan syarat-syarat kegunaan seperti berikut:

1. Tesis dan projek adalah hakmilik Universiti Teknikal Malaysia Melaka.
2. Perpustakaan Fakulti Kecerdasan Buatan dan Keselamatan Siber dibenarkan membuat salinan untuk tujuan pengajian sahaja.
3. Perpustakaan Fakulti Kecerdasan Buatan dan Keselamatan Siber dibenarkan membuat salinan tesis ini sebagai bahan pertukaran antara institusi pengajian tinggi.
4. * Sila tandakan (✓)

☒

SULIT

(Mengandungi maklumat yang berdarjah keselamatan atau kepentingan Malaysia seperti yang termaktub di dalam AKTA RAHSIA RASMI 1972)

☒

TERHAD

(Mengandungi maklumat TERHAD yang telah ditentukan oleh organisasi / badan di mana penyelidikan dijalankan)

☒

TIDAK TERHAD

(TANDATANGAN PELAJAR)

(TANDATANGAN PENYELIA)

Alamat tetap: _____

Nama Penyelia

Tarikh: _____

Tarikh: _____

CATATAN: * Jika tesis ini SULIT atau TERHAD, sila lampirkan surat daripada pihak

[TITLE OF FINAL YEAR PROJECT]

[NAME OF STUDENT]

This report is submitted in partial fulfillment of the requirements for the
Bachelor of [Computer Science (Artificial Intelligence)] with Honours.

FACULTY OF ARTIFICIAL INTELLIGENCE AND CYBER SECURITY
UNIVERSITI TEKNIKAL MALAYSIA MELAKA

[YEAR OF SUBMISSION]

DECLARATION

I hereby declare that this project report entitled

[TITLE OF FINAL YEAR PROJECT]

is written by me and is my own effort and that no part has been plagiarized
without citations.

STUDENT : _____ Date : _____
([NAME OF STUDENT])

I hereby declare that I have read this project report and found
this project report is sufficient in term of the scope and quality for the award of
Bachelor of [Computer Science (Artificial Intelligence)] with Honours.

SUPERVISOR : _____ Date : _____
([NAME OF THE SUPERVISOR])

DEDICATION

[To my beloved parents...]

ACKNOWLEDGEMENTS

[I would like to thank En. Muhammad bin Ahmad for giving assistant to complete this project successfully.....

I would also like to thank my beloved parents who have been giving me support and motivation throughout my project...]

ABSTRACT

[Abstract must start from here. Abstracts should be concise, WRITTEN IN ONE SPACING and justification and **NOT MORE THAN 300 WORDS** IN A SINGLE PAGE. Abstract is not the same as synopsis or report summary. Abstracts are WRITTEN WITH **ONE PARAGRAPH**. It should briefly state **the field of report studies, problems to be solved, solutions, research processes; and the results obtained.**]

ABSTRAK

[Abstrak mesti bermula disini. Abstrak mestilah ringkas, DITULIS DALAM SATU LANGKAU dan justifikasi **TIDAK LEBIH DARIPADA 300 PERKATAAN** DALAM SATU MUKASURAT sahaja. Abstrak tidak sama dengan sinopsis atau ringkasan tesis. Abstrak DITULIS DALAM SATU PERENGGAN. Ia hendaklah menyatakan dengan ringkas **bidang kajian tesis, masalah yang hendak diselesaikan, cara penyelesaian, proses penyelidikan; dan keputusan yang diperolehi.**]

TABLE OF CONTENTS

	PAGE
DECLARATION.....	II
DEDICATION.....	III
ACKNOWLEDGEMENTS.....	IV
ABSTRACT	V
ABSTRAK	VI
TABLE OF CONTENTS.....	VII
LIST OF TABLES	IX
LIST OF FIGURES	X
LIST OF ABBREVIATIONS	XI
LIST OF ATTACHMENTS.....	XII
CHAPTER 1: INTRODUCTION.....	13
1.1 About Chapters and First Subtitle.....	Error! Bookmark not defined.
1.1.1 Second Subtitles.....	Error! Bookmark not defined.
1.1.1.1 Third Subtitles	Error! Bookmark not defined.
CHAPTER 2: LITERATURE REVIEW AND PROJECT METHODOLOGY	14
2.1 About Figures.....	Error! Bookmark not defined.
CHAPTER 3: REQUIREMENT ANALYSIS.....	16
3.1 About Tables.....	Error! Bookmark not defined.
CHAPTER 4: DESIGN	19

4.1	About Table of Content	19
CHAPTER 5: RESULTS AND DISCUSSION		21
5.1	About List of Tables	21
5.2	About List of Figures	21
5.3	About List of Figures	21
5.4	About List of Figures	21
5.5	About List of Figures	22
CHAPTER 6: CONCLUSION.....		23
6.1	Wrap-Up	23
REFERENCES.....		24
APPENDICES		25

LIST OF TABLES**PAGE**

Table 3.1: Table Example.....	Error! Bookmark not defined.
--------------------------------------	-------------------------------------

LIST OF FIGURES

	PAGE
Figure 2.1: Caption Window	Error! Bookmark not defined.
Figure 2.2: Caption Numbering Window	Error! Bookmark not defined.
Figure 2.3: Example Figure with citation from a book (Pike, 2008. Used with permission).....	Error! Bookmark not defined.
Figure 4.1: How to update Table of Contents in this template	19
Figure 4.2: Updating the whole list.....	19

LIST OF ABBREVIATIONS

FYP - **Final Year Project**

LIST OF ATTACHMENTS

	PAGE
Appendix A	
Sample of data	19
Appendix B	
Analysis of data collection	78
.....	
.....	

CHAPTER 1: INTRODUCTION

The current landscape of road safety technology is heavily biased toward larger motor vehicles, often leaving bicycle and electric scooter riders exposed to severe infrastructural hazards. As micro-mobility solutions become increasingly integrated into urban transit networks, the vulnerability of these riders has become a critical public safety concern. A primary threat to micro-mobility users is surface-induced accidents caused by road defects such as potholes. Due to their small wheel diameters and frequently rigid suspension systems, a minor pothole that a car might easily traverse can cause an e-scooter rider to suffer an immediate loss of control or mechanical failure.

While standard GPS applications provide effective route guidance, they lack the real-time granular data necessary to warn users about immediate, localized changes in road surface quality. Furthermore, existing safety tools, such as standard dashcams, are predominantly reactive. they record accidents post-occurrence rather than preventing them. Consequently, there is an urgent demand for a proactive, real-time warning system capable of identifying hazards as they appear and assessing their specific threat level based on the rider's current state.

This study aims to bridge this safety gap by developing a proactive hazard detection framework specifically tailored to the micro-mobility community. The primary objectives are threefold: (1) to train a lightweight Convolutional Neural Network (CNN) capable of identifying potholes in real-time on a mobile device; (2) to implement a dynamic Risk Assessment Algorithm that calculates accident probability by correlating pothole dimensions with real-time vehicle velocity; and (3) to design a non-intrusive mobile interface that delivers immediate safety warnings via audio and visual cues. The scope of the proposed system focuses on a fully offline Android application utilizing TensorFlow Lite for on-device edge computing, ensuring user privacy, zero latency, and independence from network coverage.

CHAPTER 2: LITERATURE REVIEW AND PROJECT METHODOLOGY

Road hazard detection has historically relied on tactile or vibration-based sensors. Research by Kim et al. (2021) explored the use of smartphone accelerometers to detect potholes. While effective at mapping road conditions, this method is inherently reactive, identifying the hazard only after the impact has occurred. For micro-mobility users, vision-based detection is vastly superior as it facilitates proactive warnings, allowing riders to avoid hazards entirely before a collision.

The advent of Deep Learning, particularly the "You Only Look Once" (YOLO) architecture, has revolutionized real-time object detection by treating bounding box prediction as a single regression problem. Earlier iterations, such as YOLOv3 and YOLOv4, established the feasibility of road damage detection but demanded high-performance GPUs, rendering them unsuitable for mobile deployment. Recent advancements, namely YOLOv8 and YOLOv11 Nano, utilize advanced feature extraction and optimized architectures specifically designed for edge devices. These models offer a critical balance between Mean Average Precision (mAP) and inference speed, which is indispensable for real-time detection on a moving bicycle or scooter.

To achieve a "fully offline" system, on-device edge computing is facilitated through TensorFlow Lite (TFLite). This framework allows complex deep learning models to run directly on a smartphone's CPU or Neural Processing Unit (NPU). By employing weight quantization which is converting 32-bit floating-point parameters into 8-bit integers to make the model size is significantly reduced and inference latency is improved without substantial degradation in accuracy. By processing data locally, user privacy is maintained since no camera feed is uploaded to cloud servers, and the system remains robust in areas with poor cellular reception.

Despite these technological advancements, existing literature focuses almost exclusively on motor vehicles. Micro-mobility requires a specialized risk assessment model because the threat posed by a pothole is inextricably linked to the vehicle's

unique physical constraints, specifically its wheel diameter and velocity. Existing detection systems lack this critical speed-risk correlation, highlighting the necessity for the methodology proposed in this research.

CHAPTER 3: REQUIREMENT ANALYSIS

The proposed methodology follows a structured AI development lifecycle, ensuring the model is optimized for the specific computational and physical constraints of handlebar-mounted mobile devices. The framework is divided into four primary phases: Data Acquisition, Model Optimization, Algorithm Development, and System Architecture.

3.1 Data Acquisition and Pre-processing

The accuracy of the vision-based detector is heavily dependent on the quality and diversity of the training data. This project utilizes the Japan Road Damage Dataset, sourced via the Roboflow platform, to ensure the model recognizes a wide variety of pothole morphologies and asphalt conditions. To simulate real-world micro-mobility riding conditions, the dataset undergoes rigorous data augmentation. Images are subjected to motion blur to simulate vehicle vibration, brightness variations to account for diverse weather and lighting conditions, and rotational adjustments to accommodate various smartphone mounting angles on handlebars.

3.2 Model Training and Edge Optimization

The model is trained using the PyTorch framework in conjunction with the Ultralytics YOLO architecture. Training is executed on a high-performance workstation equipped with an NVIDIA RTX 3060 GPU to handle data pre-processing and CUDA-accelerated model compilation. To ensure the model meets the strict latency requirements of a mobile environment, a smaller input image size (320x320) is utilized, allowing the model to process fewer pixels and massively accelerating inference speed.

Once the desired Mean Average Precision (mAP) is achieved, the PyTorch model is exported to the tflite format. During this export process, INT8 quantization is applied to shrink the model size by approximately a factor of four. This optimization

drastically lowers RAM usage and accelerates CPU execution, enabling the model to run efficiently on standard Android smartphones (minimum 8GB RAM) without requiring an active internet connection.

3.3 Dynamic Risk Assessment Algorithm

To translate raw bounding box detections into actionable safety metrics, the system integrates a novel mathematical model to calculate the Risk Index (RI). Unlike traditional systems that treat all potholes equally, this algorithm assesses hazard severity by mathematically correlating the dimension of the pothole with the vehicle's kinetic energy:

$$RI = \left(\frac{S_{detected}}{S_{threshold}} \right) \times \left(\frac{V_{current}}{V_{limit}} \right)^2 \times 100\%$$

Where $S_{detected}$ represents the area of the pothole identified by the AI and $S_{threshold}$ acts as a calibration constant representative of the vehicle's wheel diameter. $V_{current}$ is the real-time velocity retrieved from the smartphone's GPS hardware and V_{limit} is the assumed maximum speed for the vehicle (e.g., 40 km/h). By squaring the velocity ratio, the formula perfectly mirrors the real-world physics of kinetic energy ($K.E. = \frac{1}{2} mv^2$) recognizing that the destructive impact force of a hazard scales exponentially with the rider's speed.

3.4 System Architecture and Application Development

The deployment application is developed using the Flutter framework, ensuring seamless cross-platform integration with the smartphone's camera and GPS hardware. The architecture is stratified into three main layers:

- **Input Layer:** Captures high-frequency frames from the smartphone camera and velocity data from the GPS module.
- **Inference Layer:** The TFLite interpreter processes the frames offline to detect potholes and extract bounding box coordinates at a target rate of at least 15 Frames Per Second (FPS).

- **Logic & Alert Layer:** The Risk Assessment Algorithm processes the AI output and GPS data. When the calculated Risk Index exceeds a predefined safety threshold (e.g., 75%), the system triggers immediate, non-intrusive warnings. These include a high-frequency audible "beep" and a high-contrast visual red overlay on the UI, ensuring the rider is alerted under direct sunlight with a total system latency of less than 200ms.

CHAPTER 4: DESIGN

4.1 About Table of Content

In this template, the table of contents rely on the use of styles to format chapter headings and subtitles and generates the table of content automatically. To generate the table of content in this template, place the cursor on the current table of content. Next, go to “References” tab and click on the “Update Table” icon, as shown in Figure 4.1.

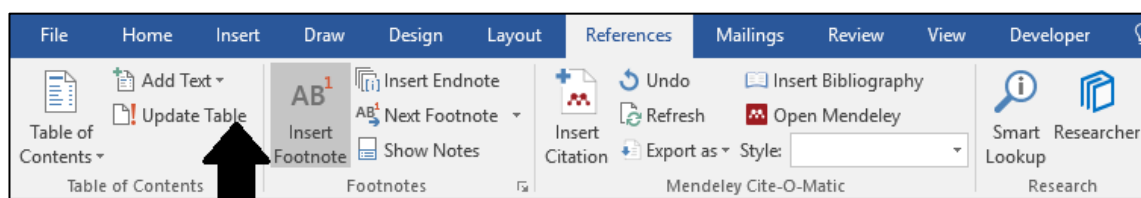


Figure 4.1: How to update Table of Contents in this template

In the “Update Table of Contents” window, choose option “Update the entire table” and click OK. Microsoft Word will generate a new table of contents based on the student’s updates.

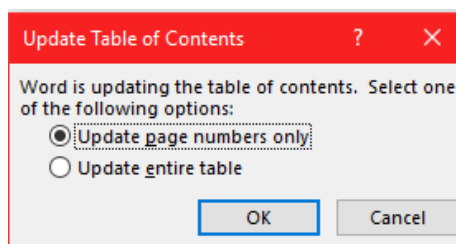


Figure 4.2: Updating the whole list

In the “Update Table of Contents” window, choose option “Update the entire table” and click OK. Microsoft Word will generate a new table of contents based on the student’s

CHAPTER 5: RESULTS AND DISCUSSION

5.1 About List of Tables

To update List of Tables, place the cursor on the list that needs to be updated. Similar to the Table of Content, click on the icon “Update Table” under References tab to list down the updates, as shown in Figure 4.1

5.2 About List of Figures

To update List of Figures, place the cursor on the list that needs to be updated. Next, click on the icon “Update Table” under References tab to list down the updates, as shown in Figure 4.1.

5.3 About List of Figures

To update List of Figures, place the cursor on the list that needs to be updated. Next, click on the icon “Update Table” under References tab to list down the updates, as shown in Figure 4.1.

5.4 About List of Figures

To update List of Figures, place the cursor on the list that needs to be updated. Next, click on the icon “Update Table” under References tab to list down the updates, as shown in Figure 4.1.

5.5 About List of Figures

To update List of Figures, place the cursor on the list that needs to be updated. Next, click on the icon “Update Table” under References tab to list down the updates, as shown in Figure 4.1.

CHAPTER 6: CONCLUSION

6.1 Wrap-Up

Please enjoy writing your FYP

REFERENCES

- Buchwalow, I. B., and Böcker, W. (2010). Immunohistochemistry: basics and methods. Berlin: Springer Verlag.
- Caamaño-Tubío, R. I., Pérez, J., Ferreiro, S., and Aldegunde, M. (2007). Peripheral serotonin dynamics in the rainbow trout (*Oncorhynchus mykiss*) *Comparative Biochemistry and Physiology Part C: Toxicology & Pharmacology*. 145(2): 245-255.
- Cameron, A. A., Plenderleith, M. B. and Snow, P. J. (1990). Organization of the spinal cord in four species of elasmobranch fishes: cytoarchitecture and distribution of serotonin and selected neuropeptides. *The Journal of Comparative Neurology*. 297: 201-218.
- Desa, M.I (1995). Bus fleet maintenance modeling in a developing country. Ph.D Thesis, University of Salford.
- Improve indigenous housing now, government told, 2007. accessed 8 February 2009, <<http://www.architecture.com.au/i-cms?page=10220>>.
- International Narcotics Control Board 1999, United Nations, accessed 1 October 1999, <<http://www.incb.org>>.
- Theusen, G. J. and Fabrycky, W. J. (1984). *Engineering Economy*. 6th. Ed. Englewood Cliffs, N. J.: Prentice Hall. 150-178.

APPENDICES